On Runs of Integers with Large Prime Factors: A Lower Bound for $k(n)$ and the Status of the Upper Bound

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Abstract

Let $k(n)$ be the maximal k such that there exists $m \le n$ with each of $m+1, \ldots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$k(n) \ \ge\ \exp\big((\log n)^{1/2-\varepsilon}\big)$$

for every fixed $\varepsilon > 0$ and all sufficiently large n, using Rankin's method and a pigeonhole argument.

We also discuss the upper bound $k(n) \le \exp((\log n)^{1/2+\varepsilon})$, which would give $\log k(n) = (\log n)^{1/2+o(1)}$. The upper bound argument requires the existence of a k_1-smooth integer in every interval $[m+1, m+k_1]$ with $m \le n$ and $k_1 = \exp((\log n)^{1/2+\varepsilon})$. We reduce this to a short-interval smooth number estimate (Hypothesis 5 below) and prove the upper bound *conditional on that hypothesis*. We discuss what the available literature (Hildebrand–Tenenbaum [1], Tenenbaum [2]) actually establishes and where the gap lies.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\le y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Set

$$\Psi(x, y) := \#\{m \le x : P^+(m) \le y\}.$$

Definition 2. A *k-run below n* is an interval $\{m+1, \ldots, m+k\}$ with $m \le n$ and $P^+(m+j) > k$ for all $1 \le j \le k$. Define $k(n)$ to be the maximum k for which a k-run below n exists.

2 The Rankin Bound

Lemma 3 (Rankin bound [2, Theorem III.5.1]). *There is an effectively computable absolute constant $A \ge 3$ such that for all $x \ge y \ge 2$ with $v := \log x / \log y \ge A$,*

$$\Psi(x, y) \;\le\; x \cdot e^{-\frac{1}{2} v \log v}.$$

Proof. Cited from [2, Theorem III.5.1]. Rankin's method with the choice $\sigma = 1 - (\log v)/\log y$ yields $\Psi(x, y) \le x\, e^{-v \log v + C_0 v / \log v}$ for an absolute constant C_0. The correction term $C_0 v / \log v \le \frac{1}{2} v \log v$ once $\log^2 v \ge 2C_0$, which holds for all $v \ge A$ with $A = e^{\sqrt{2C_0}}$. The full proof with all constants is in [2, Theorem III.5.1]. $\square$

3 The Lower Bound

Theorem 4 (Lower bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n, $k(n) \ge \exp\big((\log n)^{1/2-\varepsilon}\big)$.*

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: k_0-smooth numbers are sparse in $[1, n]$. Set $v_0 := \log n / \log k_0$. Since $\log k_0 = (\log n)^{1/2-\varepsilon}(1 + o(1))$, we get $v_0 = (\log n)^{1/2+\varepsilon}(1 + o(1)) \to \infty$. In particular $v_0 \ge A$ (the constant from Lemma 3) for all large n, so

$$\Psi(n, k_0) \;\le\; n \cdot e^{-\frac{1}{2} v_0 \log v_0}.$$

We estimate the exponent:

$$\tfrac{1}{2} v_0 \log v_0 = \tfrac{1}{2} (\log n)^{1/2+\varepsilon}(1+o(1)) \cdot (1/2+\varepsilon) \log\log n\, (1+o(1)) \;\ge\; (\log n)^{1/2+\varepsilon/2}$$

for all large n. Since $\log k_0 = (\log n)^{1/2-\varepsilon}$, this gives

$$e^{-\frac{1}{2} v_0 \log v_0} \le e^{-(\log n)^{1/2+\varepsilon/2}} \ll n^{-1} \cdot k_0^{-1},$$

so in particular $\Psi(n, k_0) \le n/(2k_0)$ for all sufficiently large n.

Step 2: Pigeonhole gives an empty block. Let $B := \lfloor n/k_0 \rfloor$. Partition $\{1, \ldots, Bk_0\}$ into B blocks $I_i := [(i-1)k_0 + 1, ik_0]$. Every k_0-smooth integer in $\{1, \ldots, n\}$ lies in one of these blocks, and

$$\Psi(n, k_0) \;\le\; \frac{n}{2k_0} \;\le\; \frac{B+1}{2} \;<\; B$$

for all $n \ge 2k_0$ (i.e. all large n). By pigeonhole, some block I_i contains no k_0-smooth integer.

Step 3: The empty block is a k_0-run. For that i, every integer in I_i has largest prime factor $> k_0$. Setting $m = (i-1)k_0 \le n$, the interval $\{m+1, \ldots, m+k_0\}$ is a k_0-run below n, so $k(n) \ge k_0 = \exp((\log n)^{1/2-\varepsilon}(1 + o(1)))$. □

4 The Upper Bound: Reduction and Gap

What needs to be proved

Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. The upper bound $k(n) \le k_1$ is equivalent to: every interval $[m+1, m+k_1]$ with $m \le n$ contains at least one k_1-smooth integer.

For $m < k_1$ this is immediate: $k_1 \in [m+1, m+k_1]$ and k_1 is k_1-smooth.

For $m \in [k_1, n]$, we need the following estimate.

Hypothesis 5 (Short-interval smooth numbers)**.** Let ρ denote the Dickman function and let $\varepsilon > 0$ be fixed. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. There exists $n_* = n_*(\varepsilon)$ such that for all $n \ge n_*$ and all integers $m \in [k_1, n]$,

$$\Psi(m + k_1,\, k_1) \;-\; \Psi(m,\, k_1) \;\ge\; 1. \tag{1}$$

(Since m is a positive integer, the difference counts exactly the k_1-smooth integers in the interval $(m, m + k_1] = \{m+1, \ldots, m+k_1\}$.)

Theorem 6 (Upper bound, conditional)**.** *Assume Hypothesis 5. For every fixed $\varepsilon > 0$ and all sufficiently large n, $k(n) \le \exp((\log n)^{1/2+\varepsilon})$.*

Proof. With Hypothesis 5 every interval $[m+1, m+k_1]$ with $m \in [k_1, n]$ contains a k_1-smooth integer, and the case $m < k_1$ was handled above. Hence no k_1-run below n exists, so $k(n) < k_1$. □

Discussion: the subtraction approach, its failure, and two routes to a proof

The main result of Hildebrand–Tenenbaum [1] is a uniform asymptotic for the *global count*. Writing $\mathcal{R}_\varepsilon = \{(x,y) : x \geq y \geq 2, \log y \geq (\log\log x)^{5/3+\varepsilon}\}$ and $\eta(x,y) := \Psi(x,y)/(x\rho(u)) - 1$ with $u = \log x/\log y$, [1, Théorème 1] gives

$$\eta(x,y) \to 0 \quad \text{uniformly for } (x,y) \in \mathcal{R}_\varepsilon \text{ as } y \to \infty. \tag{2}$$

Explicitly: for every $\delta > 0$ there exists $Y_0(\varepsilon,\delta)$ (independent of x) such that $|\eta(x,y)| \leq \delta$ for all $(x,y) \in \mathcal{R}_\varepsilon$ with $y \geq Y_0$.

A natural case split at $m = k_1^2$

The range $m \in [k_1, n]$ splits naturally at $m = k_1^2$, equivalently at $u_1 := \log m/\log k_1 = 2$.

Sub-range 1a: $m \in [k_1, k_1^2)$, i.e. $u_1 \in [1,2)$. Write $m = qk_1 + r$ with $0 \leq r < k_1$ (Euclidean division). The integer $(q+1)k_1 = m - r + k_1$ satisfies $m < (q+1)k_1 \leq m + k_1$, so $(q+1)k_1 \in [m+1, m+k_1]$. Since $m < k_1^2$ we have $q \leq k_1 - 1$, giving $q + 1 \leq k_1$. Every prime factor of $q+1$ is at most $q + 1 \leq k_1$, so $(q+1)k_1$ is k_1-smooth. Thus every interval $[m+1, m+k_1]$ with $m \in [k_1, k_1^2)$ contains a k_1-smooth integer by an elementary divisibility argument, with no analytic input.

Sub-range 1b: $m \in [k_1^2, n]$, i.e. $u_1 \geq 2$. Here we must use smooth-number asymptotics. We analyze the subtraction approach for this range.

The subtraction approach for sub-range 1b and why it fails

Subtracting (2) at $m + k_1$ and at m:

$$\Psi(m+k_1,k_1) - \Psi(m,k_1) = \underbrace{(m+k_1)\rho(u') - m\rho(u_1)}_{\text{(I)}} + \underbrace{(m+k_1)\rho(u')\eta_1 - m\rho(u_1)\eta_2}_{\text{(II)}}, \tag{3}$$

where $u' = \log(m+k_1)/\log k_1$ and η_j is $\eta(\cdot)$ at the respective argument.

Term (I) for $u_1 \geq 2$. Set $g(t) = t\rho(\log t/\log k_1)$. By the mean-value theorem, $g(m+k_1) - g(m) = k_1 g'(\xi)$ for some $\xi \in [m, m+k_1]$. Computing the derivative:

$$g'(t) = \rho(u_t)\Big(1 - \frac{\rho(u_t - 1)}{u_t\,\rho(u_t)\,\log k_1}\Big), \qquad u_t := \frac{\log t}{\log k_1}.$$

For $u_t \geq 2$, the Dickman delay-differential equation $\rho'(u) = -\rho(u-1)/u$ and the standard asymptotics [2, Corollary III.5.4] give $\rho(u-1)/\rho(u) = O(u \log u)$ for $u \geq 2$. The relative correction is therefore

$$\frac{\rho(u_t - 1)}{u_t\, \rho(u_t)\, \log k_1} = O\left(\frac{\log u_1}{\log k_1}\right) \leq \frac{(1/2 - \varepsilon) \log \log n}{(\log n)^{1/2+\varepsilon}} \to 0.$$

Hence, for sub-range 1b:

$$(\mathrm{I}) = k_1 \rho(u_1)\left(1 + O\left(\frac{\log u_1}{\log k_1}\right)\right) = k_1 \rho(u_1)(1 + o(1)). \tag{4}$$

Remark 7. The estimate (4) relies on $\rho(u_t - 1)/\rho(u_t) = O(u_t \log u_t)$, which holds for $u_t \geq 2$ but *fails* for $u_t \in (1,2)$. For $u \in (1,2)$, $\rho(u) = 1 - \log u$ and $\rho(u-1) = 1$, so $\rho(u-1)/\rho(u) = 1/(1 - \log u) \to \infty$ as $u \to 1^+$. This is why sub-range 1a requires the separate elementary argument above.

Term (II) is the fundamental obstacle. With $|\eta_1|, |\eta_2| \leq \delta$:

$$|(\mathrm{II})| \;\leq\; \big[(m + k_1)\rho(u') + m\rho(u_1)\big]\delta \;\lesssim\; 2m\rho(u_1)\delta.$$

The ratio to the main term is

$$\frac{|(\mathrm{II})|}{k_1 \rho(u_1)} \;\lesssim\; \frac{2m\delta}{k_1}.$$

For $m \in [k_1^2, n]$ this ranges from $2k_1\delta$ to $2(n/k_1)\delta$. Since $n/k_1 = \exp(\log n - (\log n)^{1/2+\varepsilon}) \to \infty$, even at the lower end of sub-range 1b the ratio is $2k_1\delta \to \infty$ for any fixed $\delta > 0$. No choice of δ depending only on ε can control this: *the error swamps the main term throughout sub-range 1b.*

Conclusion: The subtraction of two applications of (2) does not prove Hypothesis 5 for sub-range 1b.

Route 1: A Lipschitz bound on η

If one could establish, for $(x, y) \in \mathcal{R}_\varepsilon$ and $y \geq Y_1(\varepsilon)$,

$$\big|\eta(x + y, y) - \eta(x, y)\big| \;\leq\; \frac{Cy}{x \log y} \tag{5}$$

for an absolute constant C, then in (3):

$$|(\mathrm{II})| \leq (m + k_1)\rho(u')|\eta_1 - \eta_2| + k_1 \rho(u_1)|\eta_2| \leq \frac{2Ck_1\rho(u_1)}{\log k_1} + k_1 \rho(u_1)\delta.$$

Choosing $\delta = 1/4$ and $k_1 \geq e^{8C}$ (ensured by n large), Term (II) is at most $\frac{1}{2}k_1\rho(u_1)$. Since $k_1\rho(u_1) \geq 2$ (the main-term lower bound, established by the same calculation as in the lower bound proof but with k_1 in place of k_0), the difference is ≥ 1.

Bound (5) says the relative error η is Lipschitz in x with constant $O((x \log y)^{-1})$. This is the equicontinuity expected from the saddle-point analysis (the saddle point $\sigma_0 = 1 - (\log u)/\log y$ shifts by $O(1/(x \log y))$ as x increases by y), but establishing it rigorously requires a second-order expansion of $\Psi(x, y)$ in $1/\log y$ that is *uniform in x*. This is not stated in [1, 2].

Route 2: Direct Perron analysis

Rather than subtracting two global estimates, one can work with the combined Perron integral for the difference directly:

$$\Psi(m + k_1, k_1) - \Psi(m, k_1) = \frac{1}{2\pi i} \int_{\sigma_0 - iT}^{\sigma_0 + iT} F_{k_1}(s)\, \frac{(m + k_1)^s - m^s}{s}\, ds + \mathcal{E}(m, T), \tag{6}$$

where $F_y(s) = \prod_{p \leq y}(1 - p^{-s})^{-1}$, $\sigma_0 = 1 - (\log u_1)/\log k_1$, and $\mathcal{E}(m, T)$ is the tail. Expanding the kernel:

$$\frac{(m + k_1)^s - m^s}{s} = m^{s-1}k_1\left(1 + O\left(\frac{|s|k_1}{m}\right)\right). \tag{7}$$

For $m \geq k_1^2$ and s near the saddle point ($|\Im(s)| \leq C/\log m$), $|s| \approx \sigma_0 \approx 1$, so $|s|k_1/m \leq C/k_1 \to 0$. The leading-term integral $\frac{k_1}{2\pi i}\int F_{k_1}(s)m^{s-1}ds$ yields $k_1\rho(u_1)(1 + O(1/\log k_1))$ by the HT saddle-point analysis. For $m \gg k_1 \log n$ the correction $O(k_1/m)$ is $o(1)$, giving $k_1\rho(u_1)(1 + o(1)) \geq 1$ for large n. The range $m \in [k_1^2, k_1 \log n \cdot k_1] = [k_1^2, k_1^2 \log n]$ is covered by the same argument (the correction is $O(k_1/k_1^2) = O(1/k_1) = o(1)$ for $m \geq k_1^2$).

This sketch glosses over three genuine technical obstacles, which a complete proof would need to address.

Obstacle 1 (Bounding the correction-term integral). The correction $O(|s|k_1/m)$ in (7) contributes $O(k_1^2/m)$ times the integral $\int_{\sigma_0 - iT}^{\sigma_0 + iT} |s||F_{k_1}(s)|m^{\sigma_0 - 1}|ds|$. This s-weighted integral is *not* a direct consequence of the Perron formula for $\Psi(m, k_1)$: inserting an extra factor $|s|^2/m$ into the Perron integrand requires bounding $|F_{k_1}(\sigma_0 + it)|$ for $|t|$ up to T, and such vertical-strip bounds must be stated explicitly.

Obstacle 2 (Tail uniformity in m). The tail satisfies

$$\mathcal{E}(m,T) = O\left(\frac{(m+k_1)^{\sigma_0} F_{k_1}(\sigma_0)}{T}\right) \approx O\left(\frac{m\rho(u_1)}{T}\right).$$

To achieve $\mathcal{E}(m,T) = o(k_1\rho(u_1))$ one needs $T \gg m/k_1$. Since $\sigma_0 = 1 - (\log u_1)/\log k_1$ depends on m, both the choice of T and the resulting error must be controlled *uniformly* over all $m \in [k_1^2, n]$. As m/k_1 ranges up to $n/k_1 \to \infty$, this requires the saddle-point analysis to remain valid for T growing without bound, which is a non-trivial uniformity statement.

Obstacle 3 (Contour validity for large $|\Im(s)|$). The expansion (7) and the bound on $F_{k_1}(\sigma_0 + it)$ must hold uniformly for $|\Im(s)|$ up to $T = O(m/k_1) \to \infty$. In the HT framework, $|F_y(\sigma_0 + it)| \ll |F_y(\sigma_0)|e^{-c(\log y)|t|}$ for some $c > 0$ when $|t| \leq 1$; extending this to $|t| \to \infty$ as required here demands additional input on the distribution of $F_y(s)$ on tall rectangles, which is available (it follows from zero-free regions for $F_y(s)$) but must be made explicit.

Remark 8 (Summary: what an unconditional proof requires)**.** Both routes are refinements of the HT saddle-point method, not new ideas. Route 1 requires a second-order uniform expansion $\Psi(x,y) = x\rho(u)(1 + c_1(u)/\log y + O((\log y)^{-2}))$ with $c_1(u)$ satisfying a Lipschitz bound, uniform in x. Route 2 requires resolving Obstacles 1–3 above: bounding s-weighted Perron integrals, controlling tails uniformly in m, and extending saddle-point bounds to tall contours. All necessary inputs—the Perron integral for $F_y(s)m^s$, bounds on $|F_y(\sigma + it)|$, the Dickman function and its derivatives—are available in principle via [1, 2], but the arguments have not been assembled into a complete proof of Hypothesis 5.

5 Main Result

Theorem 9. (a) *(Unconditional) For every fixed $\varepsilon > 0$ and all large n: $k(n) \geq \exp((\log n)^{1/2-\varepsilon})$.*

(b) *(Conditional on Hypothesis 5) For every fixed $\varepsilon > 0$ and all large n: $k(n) \leq \exp((\log n)^{1/2+\varepsilon})$.*

Hence, conditional on Hypothesis 5, $\log k(n) = (\log n)^{1/2+o(1)}$. Hypothesis 5 is a short-interval smooth number estimate plausibly within reach of the Hildebrand–Tenenbaum method but not verified directly from the published sources [1, 2] here.

Proof. Part (a) is Theorem 4. Part (b) is Theorem 6. □

References

[1] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.

[2] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 3rd ed., 2015.